

CTNT 2026 - Elliptic Curve Cryptography
Suggested Exercises

1. Find the least non-negative residue class of $6^{182} \bmod 13$.
2. Find the least non-negative residue class of $6^{-58} \bmod 13$.
3. Show that 6 is a primitive root modulo 13.
4. Find x such that $6^x \equiv 5 \bmod 13$.
5. Find the multiplicative order of 2 modulo 4003.
6. Make a table of primes (up to a bound) for which 2 is a primitive root. What kind of statistics do you notice?
7. Find a natural number $b \geq 1$ such that $65^b \equiv 2 \bmod 4003$.
8. Show that $y = x^3$ is not an elliptic curve.
9. Let $E/\mathbb{Q}$ be the elliptic curve $y^2 = x^3 - 2$ and let $P = (3, 5)$. Use the geometric definition of the addition on E to find the coordinates of $2P$ and $3P$.
10. Let E be the elliptic curve given by $y^2 = x^3 + 1$ over $\mathbb{F}_7$. Find all the points on E over $\mathbb{F}_7$ and describe the group structure of $E(\mathbb{F}_7)$.
11. Let p be a prime and let $E : y^2 = x^3 + 4x + 6$.
 - (a) For what primes p is the curve E an elliptic curve?
 - (b) For $p = 3, 5, 7, 11, 13, 17, 19, 4001, 4003$, find the number of points on $E(\mathbb{F}_p)$, the difference $(p + 1) - \#E(\mathbb{F}_p)$, and $2\sqrt{p}$. Verify that the Hasse bound holds in each case.
 - (c) Show that $E(\mathbb{F}_{4001})$ is a cyclic group.
12. Use Shank's baby-step giant-step algorithm to find x such that

$$5^x \equiv 3 \bmod 23.$$
 - (a) The order of 5 mod 23 is 22, so compute $n = 1 + \lfloor \sqrt{22} \rfloor$.
 - (b) Let $g = 5$. Compute List 1: $1, g, g^2, \dots, g^n \bmod 23$.
 - (c) Compute $u \equiv g^{-n} \bmod 23$.
 - (d) Let $h = 3$. Compute List 2: $h, h \cdot u, h \cdot u^2, \dots, h \cdot u^n \bmod 23$.
 - (e) Use the information above to finish the baby-step giant-step algorithm and find x such that $5^x \equiv 3 \bmod 23$.
13. Prove that Shank's baby-step giant-step algorithm works, i.e., show that (i) there is always a coincidence in List 1 and List 2, and (ii) that $x = i + j \cdot n$ is a number such that $g^x \equiv h \bmod p$.
14. Let $p = 17389$, let $g = 9704$, and let $h = 13896$. Use Shank's algorithm to find x such that $g^x \equiv h \bmod p$.

15. Let $p = 73$, let $E : y^2 = x^3 + 8x + 7$ over $\mathbb{F}_p$, and let $P = (32, 53)$ and $Q = (35, 47)$. Use a probabilistic collision algorithm to find n such that $n \cdot P = Q$.
16. Let $p = 73$, let $E : y^2 = x^3 + 8x + 7$ over $\mathbb{F}_p$. Determine the group structure of $E(\mathbb{F}_{73})$.
17. Alice and Bob want to communicate using a Diffie–Hellman key exchange first, using $p = 13$ and $g = 6$.
- Alice picks $a = 2$. Compute $A \equiv g^a \pmod{p}$.
 - Alice receives $B = 2$ from Bob. Compute the secret key produced by the Diffie–Hellman key exchange that Alice and Bob will share.
 - Your name is Eve and you are *very nosy*. You intercept some messages from Alice and Bob who are setting up a Diffie–Hellman key exchange. You now know that $p = 13$, $g = 6$, and intercept $A = 3$ from Alice, and $B = 5$ from Bob. Use this information to generate the secret key that Alice and Bob are planning to share. (Note: it is not the same key that they were sharing in part (b) of this problem).
18. The following is a table of some powers modulo 17:

$x \pmod{17}$	x^2	x^3	x^4	x^5	x^6	x^7	x^8	x^9	x^{10}	x^{11}	x^{12}	x^{13}	x^{14}	x^{15}	x^{16}
4	16	13	1	4	16	13	1	4	16	13	1	4	16	13	1
6	2	12	4	7	8	14	16	11	15	5	13	10	9	3	1
9	13	15	16	8	4	2	1	9	13	15	16	8	4	2	1
10	15	14	4	6	9	5	16	7	2	3	13	11	8	12	1
13	16	4	1	13	16	4	1	13	16	4	1	13	16	4	1
15	4	9	16	2	13	8	1	15	4	9	16	2	13	8	1

- Princess Leia is expecting communications from the rebel alliance, and sets up an Elgamal public cryptography system, with $p = 17$ and $g = 10$. She chooses $a = 5$ and publishes p , g , and $A \equiv g^a \pmod{p}$.
 - Compute A .
 - Luke wants to send the message $m = 3$ to Leia. Explain what steps Luke has to do to send the message, and produce an encrypted version (c_1, c_2) of the message.
 - Leia receives an encrypted message $(15, 9)$ from Chewbacca. Use the Elgamal method to decipher Chewbacca's message.
 - Governor Tarkin intercepts a rebel transmission $(13, 2)$, which has been encoded with an Elgamal cryptosystem (different from the one above). He knows that $p = 17$, $g = 6$, and $A = 9$. Decipher the transmission.
19. The number $p = 179424673$ is prime and 15 is a primitive root mod p . Find x such that $15^x \equiv 2 \pmod{p}$.
20. (Euler's Theorem 2.0) Let p, q be two distinct primes, and let $d = \gcd(p-1, q-1)$. Let a be an integer such that $\gcd(a, pq) = 1$. Show that $a^{(p-1)(q-1)/d} \equiv 1 \pmod{pq}$.
21. Let $p > 2$ be a prime, let e be an integer relatively prime to $p-1$, and let $d \equiv e^{-1} \pmod{p}$. Let c be an integer. Show that the equation $x^e \equiv c \pmod{p}$ has a unique solution $x \equiv c^d \pmod{p}$.

22. Let p, q be two distinct odd primes, and let e be an integer with $\gcd(e, (p-1)(q-1)) = 1$. Let $d \equiv e^{-1} \pmod{(p-1)(q-1)}$. Show that the equation $x^e \equiv c \pmod{pq}$ has a unique solution $x \equiv c^d \pmod{pq}$.
23. Find a prime number p such that $10^{30} < p < 10^{40}$.
24. Let $N = 1147$.
- Find a factorization into prime powers of $N - 1 = 1146$.
 - The multiplicative order of 2 modulo N is 180. Use this fact to show that N is not a prime number.
 - Is 2 a Fermat witness for the compositeness of N ?
 - Is 2 a Miller-Rabin witness for the compositeness of N ?
 - Use Pollard's $p - 1$ method to find a factorization of N .
25. In this problem, $N = 1147$. Alice wants to set up an RSA encryption scheme with N as her modulus.
- Can Alice choose $e = 3$ as an encryption exponent? If so, find the corresponding decryption exponent d . If not, explain why not.
 - Can Alice choose $e = 7$ as an encryption exponent? If so, find the corresponding decryption exponent d . If not, explain why not.
 - Alice publishes $(N, e) = (1147, 7)$ for her RSA encryption. Bob wants to send the message "HI" to Alice. Each letter is encoded $A = 01, B = 02, \dots$ and messages are encoded in message blocks of two letters (so "YOYO" would be encoded as 2515 2515).
 - Encode "HI" into numeric form, and encrypt it using RSA with $(N, e) = (1147, 7)$.
 - Alice receives the message 793 from Bob. Explain how Alice would decrypt it.
 - Decrypt the message 793 into an actual English word. (Note: the plaintext message "0868" makes no sense because "68" does not correspond to a letter. However, $0868 \equiv 2015 \pmod{1147}$ and "2015" corresponds to the word "TO".)
26. (a) Consider the number $n = 57091$, which was chosen at random in the interval $[1, 58000]$. According to the prime number theorem, how likely is it that n is a prime number?
- (b) Consider the numbers 57091, 57093, 57097, and 57099. If we only take the prime number theorem into consideration, what is the probability that all four numbers are composite?
- (c) Check that the number $a = 2$ is a Miller-Rabin witness for the compositeness of 57091, 57093, and 57099. Are they composite numbers? Why?
- (d) We checked $a = 2$ in the Miller-Rabin test for $n = 57097$ and found the following:
- $$2^{7137} \equiv 9149 \pmod{n}, \quad 2^{2 \cdot 7137} \equiv 57096 \pmod{n}, \quad \text{and} \quad 2^{4 \cdot 7137} \equiv 1 \pmod{n}.$$
- First, verify that we are using the Miller-Rabin test correctly. Then, can you conclude that n is prime? If not, what is your confidence that 57097 is prime?
- (e) We checked $a = 37$ in the Miller-Rabin test for $n = 57097$ and found the following:
- $$37^{7137} \equiv 12298 \pmod{n}, \quad 37^{2 \cdot 7137} \equiv 47948 \pmod{n}, \quad \text{and} \quad 37^{4 \cdot 7137} \equiv 57096 \pmod{n}.$$
- Can you conclude that n is prime? If not, what is your confidence now that 57097 is prime?

27. The following is a table of the powers of 3 modulo 43.

$x \bmod 43$	x^2	x^3	x^4	x^5	x^6	x^7	x^8	x^9	x^{10}	x^{11}	x^{12}	x^{13}
3	9	27	38	28	41	37	25	32	10	30	4	12
	x^{14}	x^{15}	x^{16}	x^{17}	x^{18}	x^{19}	x^{20}	x^{21}	x^{22}	x^{23}	x^{24}	x^{25}
	36	22	23	26	35	19	14	42	40	34	16	5
	x^{26}	x^{27}	x^{28}	x^{29}	x^{30}	x^{31}	x^{32}	x^{33}	x^{34}	x^{35}	x^{36}	x^{37}
	15	2	6	18	11	33	13	39	31	7	21	20
	x^{38}	x^{39}	x^{40}	x^{41}	x^{42}							
	17	8	24	29	1							

- Find an element $g \bmod 43$ such that the multiplicative order of g is 7.
- Samantha is setting up a DSA digital signature with $p = 43$, $q = 7$, and the element g you found in part (1). She chooses a secret exponent $a = 2$. Compute $A \equiv g^2 \bmod p$.
- Samantha's document is $D \equiv 5 \bmod 7$. She chooses $k = 3$. Compute the signature (S_1, S_2) of Samantha's document using the DSA digital signature system.
- Victor receives a document $D \equiv 4 \bmod 7$ and a DSA signature $(37, 6)$. Is this a valid DSA signature for the document D , with a DSA set up as above?
- What would Samantha's signature be if she signs a document $D \equiv 5 \bmod 43$, using an Elgamal digital signature for $p = 43$ and $g = 3$, with $a = 2$ and $k = 5$?

28. Find a factorization of

179694915979410667329161284495732461563675618080126000
708889188355317264603414909334933722478686507552308558
641999292218144366847228740520652579374956943483892631
711525225256544109808191706117425097024407180103648316
38288518852689.

29. Decipher the following code that was written with an RSA encryption scheme, with $(N, e) = (3127, 3)$:

1732 174 571 2100 1732 174 1498 1701 1135 1269 2673 1498 3118
174 2673 2673 2576 1433 174 1498 2576 2375 1732 1498 2576 571
174 1498 1701 1135 1269 2673 1498 174 2913 2576 3118.

Note: the plaintext message was encoded letter by letter using their ASCII code, i.e., $A = 065$, $B = 066$, etc., and each encoded letter was used as a message block that was, in turn, encrypted using RSA. For example, the letter A would be encoded as 065, and it would be encrypted as $c \equiv 65^3 \equiv 2576 \bmod 3127$.

30. Let E be the elliptic curve $y^2 = x^3 + x + 1$ defined over the field $\mathbb{F}_5$. Let $P = (0, 1)$ and $Q = (2, 4)$. Use the formulas of addition on an elliptic curve to compute the point $P \oplus Q$.

31. Let E be the elliptic curve $y^2 = x^3 + x + 1$ defined over the field $\mathbb{F}_5$. Let $Q = (2, 4)$. Use the formulas of addition to compute the point $2Q$.
32. Use Hasse's theorem to give an upper bound for the total number of points on $E(\mathbb{F}_5)$, where E is the elliptic curve $y^2 = x^3 + x + 1$ defined over the field $\mathbb{F}_5$.
33. Find all the points on $E(\mathbb{F}_5)$, where E is $y^2 = x^3 + x + 1$ over the field $\mathbb{F}_5$.
34. Let $p = 541$ (which is a prime), and let E be the elliptic curve $y^2 = x^3 + x + 1$ defined over $\mathbb{F}_{541}$. Let $P = (72, 70)$ on $E(\mathbb{F}_{541})$. The following is a list of the multiples $n \cdot P$ for $1 \leq n \leq 59$, in order:

$(72, 70), (424, 71), (9, 110), (338, 159), (255, 123),$
 $(161, 528), (147, 468), (168, 416), (480, 353), (454, 92),$
 $(360, 174), (264, 41), (152, 438), (468, 56), (437, 44),$
 $(68, 447), (459, 293), (115, 326), (328, 507), (278, 318),$
 $(113, 117), (534, 456), (307, 277), (1, 57, 1), (491, 440),$
 $(107, 249), (465, 115), (67, 517), (301, 61), (301, 480),$
 $(67, 24), (465, 426), (107, 292), (491, 101), (1, 484),$
 $(307, 264), (534, 85), (113, 424), (278, 223), (328, 34),$
 $(115, 215), (459, 248), (68, 94), (437, 497), (468, 485),$
 $(152, 103), (264, 500), (360, 367, 1), (454, 449), (480, 188),$
 $(168, 125), (147, 73), (161, 13), (255, 418), (338, 382),$
 $(9, 431), (424, 470), (72, 471), (0 : 1 : 0),$

where $(0 : 1 : 0)$ is $\mathcal{O}$, the point at infinity. In other words, $P = (72, 70)$, $2P = (424, 71)$, $3P = (9, 110)$, $\dots$, $6P = (161, 528)$, etc.

- (a) Explain why the order of P in $E(\mathbb{F}_{541})$ is 59.
 - (b) Alice and Bob want to set up an elliptic Diffie-Hellman key exchange with $p = 541$ and E and P as above. Alice chooses $n_A = 10$ as her secret integer. What point Q_A should Alice send to Bob?
 - (c) Alice and Bob want to setup an elliptic Diffie-Hellman key exchange with $p = 541$ and E and P as above. Alice chooses $n_A = 10$ as her secret integer, and she receives $Q_B = (459, 293)$ from Bob. Determine the secret point that is shared between Alice and Bob.
 - (d) Eve intercepts a communication between Alice and Bob (not the one above). Eve now knows that Alice sent $Q_A = (534, 456)$ to Bob, and Bob sent $Q_B = (255, 123)$ to Alice. Determine the secret point that now Alice and Bob share.
35. In this exercise $p = 541$ and E and P are as in the previous problem.
- (a) Alice sets up an elliptic Elgamal cryptosystem with $p = 541$ and E and P as in Problem 3. She chooses a private key $n_A = 13$, and publishes $Q_A = 13P = (152, 438)$. Bob wants to send a plaintext $M = (454, 449)$ to Alice. Please help Bob by encrypting his message into a ciphertext (C_1, C_2) to be sent to Alice.
 - (b) Using the same elliptic Elgamal system as in part (1), Alice receives a message $(C_1, C_2) = ((9, 110), (72, 70))$ from Bob. What was the plaintext that Bob encrypted?

- (c) Samantha wants to create an elliptic curve digital signature using $p = 541$, E and $G = P = (72, 70)$ as in Problem 3, and $q = 59$. Her secret signing key is $s = 5$.
- Compute $V = s \cdot G$.
 - She chooses a document $d \equiv 17 \pmod{59}$, and chooses a random element $e \equiv 3 \pmod{59}$. Compute $e \cdot G$.
 - Compute the signature $(d, (s_1, s_2))$ for $d \equiv 17 \pmod{59}$ using $e = 3$.
36. In this problem, $N = 221$.
- Is $a = 2$ a Miller-Rabin witness for $N = 221$?
 - We have $3^{55} \equiv 198 \pmod{221}$ and $3^{110} \equiv 87 \pmod{221}$. Use this information and the Miller-Rabin test to determine whether $N = 221$ is a composite number.
 - Use Pollard's $p - 1$ factoring algorithm to find a factorization of $N = 221$ as a product of prime numbers.
37. Let $N = 143$ and $E : y^2 = x^3 + x - 1$ and $P = (1, 1)$. Use Lenstra's algorithm to factor N , using E and P modulo N .